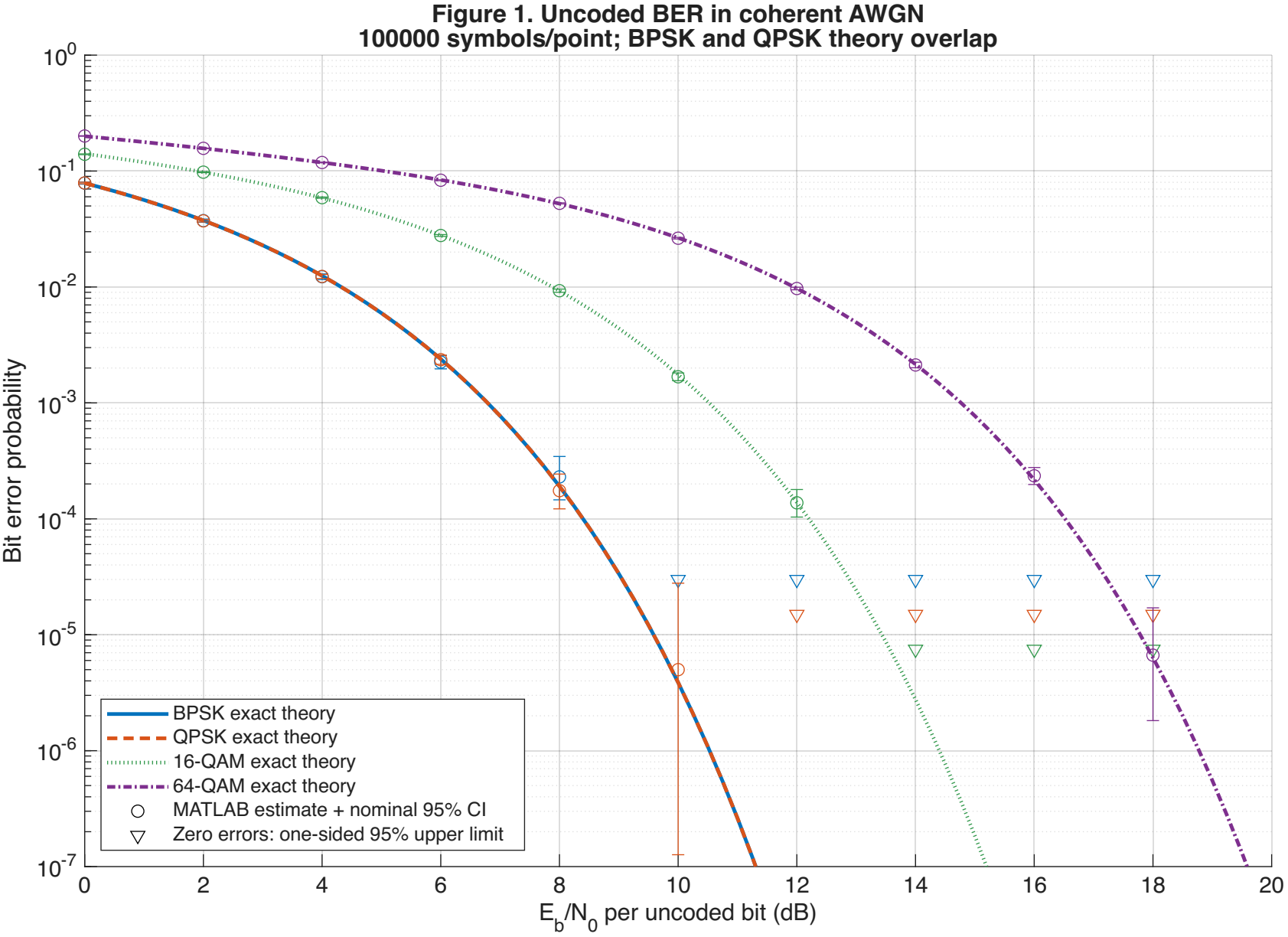

Figure 1. Uncoded BER in coherent AWGN
100000 symbols/point; BPSK and QPSK theory overlap
Bit error probability
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0
2
4
6
8
10
12
14
16
18
20
E_b/N_0 per uncoded bit (dB)
BPSK exact theory
QPSK exact theory
16-QAM exact theory
64-QAM exact theory
MATLAB estimate + nominal 95% CI
Zero errors: one-sided 95% upper limit